

Mental imagery, fascia and movement

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Introduction

Cognition (the neural processes involved in perceiving, interpreting and using sensory information) is an integral component of movement. Cognitive elements of awareness, consciousness, somatics and mindfulness have traditionally referred to an individual's power to strategically use the mind to influence physical and emotional spheres. Cognition is important to daily functioning, mind-body and movement practices, sports and rehabilitation. As sensory stimuli from the environment often require split-second motor responses, the body and mind work together to make predictions and take appropriate actions. Cognitive tools of mentally envisioning and rehearsing scenes or outcomes are known as mental imagery (MI) (Guillot and Collet, 2010). Such tools are particularly relevant in sport and movement, for which a combination of mental and physical elements is fundamental. Research into the potential associations between cognition, including MI, and fascia, is in its infancy. However, based on research to date, there are at least three possible avenues in which a pairing of MI with fascia can be identified to support movement and well-being: (1) Fascia can serve as an attentional focus, including during MI; (2) MI positively affects muscle tissue (Slimani et al., 2016) and, given the cellular, physiological and functional similarities existing between muscle and fascia tissue, MI likely also affects fascia; and (3) fascia can respond to

chemical and mechanical stimuli (Schleip et al., 2019), so its responsiveness to cognitive stimuli such as MI cannot be ruled out.

This chapter presents a rationale for possible interactions between MI and fascia, and discusses ways in which the two can communicate to support mental and physical well-being. Specifically, examples are provided of traditional and novel ways in which fascia-related MI can assist the mind-body synergy within a context of movement, sports and rehabilitation.

What is mental imagery?

MI is the self-generated cognitive process of creating any picture, experience (visual, kinesthetic, auditory, gustatory, etc.) or scene in the mind, typically with the goal of influencing a physical or psychological outcome (Guillot and Collet, 2010). The closer the mental image is to creating a multisensory experience of realism, the better and more effective MI is. For example, sitting in an armchair at home, one can image being at the beach with feet resting on warm sand (kinesthetic), watching the sun dipping below the horizon (visual), hearing the sound of the waves breaking on the shore (auditory), and tasting the salt air (gustatory), all in order to bring the imaged experience closer and closer to that of actually being there.

As a fundamentally trainable human skill, the ability to easily and clearly create a mental

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image gets better with practice, even in cases where cognition is negatively impacted, such as in sport-related concussions or neurodegenerative diseases (Abraham et al., 2018). When associated with movement, MI is called motor imagery, which can be used solely or in conjunction with actual physical execution of the movement. For example, while imaging bending and extending the elbow, a person can focus on the feeling (kinesthetic) and/or visual experience, with or without actually moving it. The details of a real and/or an imaged experience can arise as the performer notices his or her internal kinesthetic experiences of the body, movement, sensations and thoughts, as well as details of the imaged environment's scenes and objects. Visual MI can take the perspective of an internal (first-person) imaging of the scene through one's own eyes or an external (third-person) view of oneself in a scene from some distance away.

Mental images can include both real (anatomical or biomechanical) or fictional (metaphorical) contents about the body itself, including its tissues, cells, organs and body parts. Metaphorical images are those that adapt a borrowed quality/idea from one object and apply it to another. For example, if while landing from a jump one can image the foot to be an elastic arched rubber band that is being slightly flattened on impact, the quality and characteristics of the metaphor (the rubber band's elasticity and energy absorption and storage capabilities) may facilitate the foot's normal biomechanics and thus possibly enhance the mobility and flexibility required for maximum performance and injury prevention.

Beneficial effects of MI include increased muscle power, range of motion and movement efficiency; improved memory and dual tasking; and increased self-confidence and reduced anxiety and pain (motor, cognitive and psychological

outcomes, respectively). Such diverse beneficial effects explain why MI is used as a training method to enhance performance in sports, movement and rehabilitation (Dickstein and Deutsch, 2007; Abraham et al., 2016, 2017). As a training method, MI can be used in isolation or in conjunction with other physical training approaches, including physical and occupational therapy, Feldenkrais, yoga and Pilates. Furthermore, MI has assisted with movement outcomes in a variety of populations, from athletes and dancers to older adults and people with neurodegenerative conditions, such as Parkinson's disease and multiple sclerosis. Indeed, MI is available even when pain, paralysis or physical confinement limits or prohibits physical training, such as after an injury or surgery.

MI assists beginners and elite experts at all stages of learning a task by cueing attention to the body, for example, by saying, "As you move, image your entire body filled with balloons" (Figure 47.1) or by offering a reassuring image to overcome fear or other limitations in range of motion. MI's advantages include its ready availability, low cost and safety of use.

How does mental imagery work?

How and why MI positively affects physiological and psychological aspects of movement is not yet fully understood. It is known that mentally imaging a motor task (e.g. lifting a leg) activates the same brain regions as those responsible for actually performing the task, albeit to a somewhat more limited degree (Guillot and Collet, 2010). Such brain activation provides a training effect without physical execution. Furthermore, mentally replicating a motor task, along with other task-related sensory and cognitive information, allows for further improvements in the performance of the task.

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Figure 47.1 Cueing attention to the body through MI's metaphor of moving while imaging one's entire body filled with balloons

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MI's contribution to neural changes in the central nervous system results in greater relaxation and improved programming of the motor system (Yue and Cole, 1992; Yue et al., 1996). Given the close anatomical and functional association between muscle and fascia tissues, neural processes within the myofascial system can contribute to MI's physiological and psychological mechanisms of effect. For example, by addressing

both muscle sarcomeres and their surrounding fascia, MI could intensify the recruitment of motor units (Helm et al., 2015). Insofar as MI addresses the autonomic nervous system, blood flow and other physiological variables, such as heart rate and stress levels, are impacted (Collet et al., 2013).

Potential psychological mechanisms of MI include facilitation of the cognitive elements

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associated with skill execution and motor learning, including attentional focus, self-awareness and various movement patterns. As MI directs attentional focus to the engaged fascial tissue throughout movement preparation and execution, fascia's role in skill execution and motor learning can interact with, and potentially be enhanced by, the cognitive tool of MI for mutual benefit. As MI supports healthy functional fascial tissue, movement and performance are improved. In return, as the body feels and functions better, one more easily imagines positive outcomes and so is able to use MI more effectively. Details of the ways in which this beneficial interaction could happen are included below.

The role of sensory information in mental imagery

Previous experiences play a part in perceiving, interpreting and responding to subsequent ones, as in conscious perception where memories or experiences become integrated with the ongoing sensory information. Similarly, personal experiences and sensory information can affect MI. That is, having a previous experience with a motor task, scene or sensation often promotes a more vivid, accurate and complete mental image created by the brain more quickly and easily. A previous experience that entails wrong or negative information and memories about the task tends to hamper both physical performance and MI. For example, people experiencing pain are often limited in their physical capabilities and so are prone to generate more negative mental images. Individuals with chronic pain experience greater difficulties engaging in MI when compared with asymptomatic individuals, including more time being required to complete MI tasks and less clarity of the mental images being formed (Bowering et al., 2013). It is logical to assume that pain originating from fascial tissue (such as in

low back pain or fascia-related syndromes) will also impede MI. Interestingly, MI can reduce, or increase, pain depending on the imaged contents (Fardo et al., 2015). Facilitating more positive mental images about the fascial tissue could, therefore, offer a therapeutic tool for sufferers of fascia-related pain.

Another way in which sensory information from fascia can affect MI is through its effects on body schema. Body schema is the mental representation of one's body and its parts in space and in relation to each other. Body schema is formed from sensory information derived from the different body parts and tissues (often referred to as proprioception). Body schema deficits, as seen in brain damage or neuro-degenerative diseases, degrades MI. Insofar as the sensory and mechanical capabilities of fascial tissue involved in biomechanics, posture and movement contribute to body schema, sensory-proprioceptive information from fascia can affect MI, including that of body schema. In a study of people with Parkinson's disease, participants were able to improve their pelvic schema (assessed by self-drawings) following a 2-week MI training intervention (Abraham et al., 2019b). An improvement in body schema due to MI training can be, at least in part, attributed to clearer and intensified fascia-related sensory-proprioceptive information leading to increased fascial awareness. More ways in which MI can affect sensory and motor experiences are discussed below.

Can mental imagery modulate actual motor and sensory experiences?

As MI uses, and is affected by, sensory and motor experiences and information that includes those from the fascia, the path from MI to fascia-related sensory-motor experiences opens the door to fascia-related cognitive-motor integration. MI's facilitation of correct body biomechanics and

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schema enhances motor and non-motor aspects of performance. By directing attentional focus and facilitating the many components associated with a task, MI helps shape the mental experience of that motor task. Adding movement to MI further promotes better and safer movement as increased brain activity facilitates a direct, immediate and concrete connection between the mental image and its desired effect on movement, posture, body schema and emotional well-being. In a study in dancers, MI ability and MI “positive” contents were positively correlated with increased confidence and decreased anxiety levels, suggesting that MI plays a role in psychological aspects of performance (Monsma and Overby, 2004).

In summary, MI is affected by motor and sensory experiences and information from the body and the environment. MI is also capable of facilitating and modulating sensory and motor actual experiences and information.

Mental imagery of fascia

Mental imagery of fascia (aka fascial MI) specifically addresses the dynamic structure, range of function, and applicable metaphors of fascial tissue, given that the latter is often not easily felt, palpated, or recognized. One fascial MI approach currently practiced is a division of Dynamic

Neuro-Cognitive Imagery (DNI®; also known as “The Franklin Method”®). DNI is an evidence- and science-based (Heiland et al., 2012, 2013; Franklin, 2014; Abraham et al., 2019a) MI approach to movement, postural retraining and rehabilitation that combines MI components (e.g. metaphorical imagery, anatomical imagery, and visual and kinesthetic imagery) with anatomical and biomechanical embodiment and mindful movement. DNI’s tools of self-talk and self-touch increase fascial understanding and awareness through experiential embodiment. DNI’s fascial MI combines anatomical, biomechanical and metaphorical cues with visual and kinesthetic MI of the dynamic, three-dimensional nature of the movement’s fascial actions. For example, the thoraco-lumbar fascia can be imaged as a slippery web changing its shape as the arm reaches forward and up (Figure 47.2). Given the dynamic nature of fascia and its contribution to biomechanics and movement, fascial MI used alongside actual movement in training and rehabilitative settings can later be used by the performer and the trainer/therapist to communicate the desired aspects of the movement. For example, the expression “thoracolumbar fascia as a web” (Figure 2) could serve as a means of communication between the performer and their trainer/therapist to provide a clear and efficient cue in subsequent interactions.

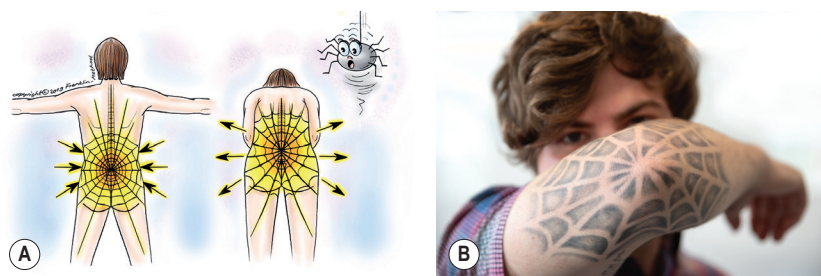


Figure 47.2

(A) Metaphorical DNI of the thoracolumbar fascia as a moveable web (drawn by Mr. Eric Franklin, all rights reserved). (B) A tattoo of a web on a skin illustrates the dynamic nature of the web (photographer: Karina Glikin-Yaari, all rights reserved).

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Using mental imagery as a cueing and instructional strategy

Integrating MI within an exercise and training regimen involving fascia is a process of combining art and science from the fields of cognition, communication and psychology. Communicating MI with cues and instructions in movement practice requires concentration, self-awareness and attentional focus. It also requires that the cues be tailored to the knowledge, skills and experiences of the performer. Anatomical and biomechanical MI especially require a knowledge and vocabulary of the field. Metaphorical MI is personal and subjective, so such cues need to be adjusted and tailored to the specific individual. For example, cueing a client to relax the shoulders with the DNI image of “the fascia of your upper trapezius melting like honey” may unexpectedly lead to tense shoulders in an individual who is allergic to honey. Therefore, creating efficient and effective MI requires ongoing mutual feedback between the instructor/therapist and the performer. It is important to regularly ask the performer to notice and report any thoughts about the experiences with different mental images. Doing so allows the brain to register and record the novel mental image and the experience associated with it. Additionally, asking for feedback from the performer allows for a clear and common MI language to be built with the goal of optimizing motor performance. With this caveat in mind, we invite you to “try on, like trying on different outfits” the DNI images below. Notice for yourself any changes in your physical and cognitive experiences of movement.

Dynamic neuro-cognitive imagery exercises for mental imagery and fascia

Below are three examples that specifically address fascial tissue with fascial DNI. Each

consists of anatomical, biomechanical and metaphorical MI, although to different degrees, and all three are performed while moving.

Exercise 1. Reciprocal motion of the fascia of the arm (Figure 47.3)

1. Focus on the muscles of the right upper arm and image the fascia located there surrounding, separating and connecting the muscles (brachialis, biceps, triceps).
2. Image the fascia to be a stocking, with a color and a texture of your choosing.
3. Flex your elbow, and as the biceps contracts, image the stocking around it shortening and widening.
4. Image the biceps muscle bulging and filling the fascial stocking, as the fascial stocking is expanding and creating space for the biceps muscle (Figure 47. 3B).
5. Image the triceps stocking stretching, lengthening and narrowing as the elbow flexes.
6. Image the fascia “squishing the triceps into length, like squishing toothpaste down its tube”.
7. As you extend the elbow, image the opposite: the triceps muscle is bulging and filling the fascial stocking and the fascial stocking expanding and pushing back against the triceps muscle.
8. At the same time, image the biceps stocking stretching, lengthening and narrowing as the elbow extends. Image the fascia squishing the biceps into length, like squishing toothpaste down its tube (Figure 47. 3A).
9. Repeat 5–6 times while focusing on this mental image during the movement.
10. Let go of the hands and compare the right and the left side. Do you feel any

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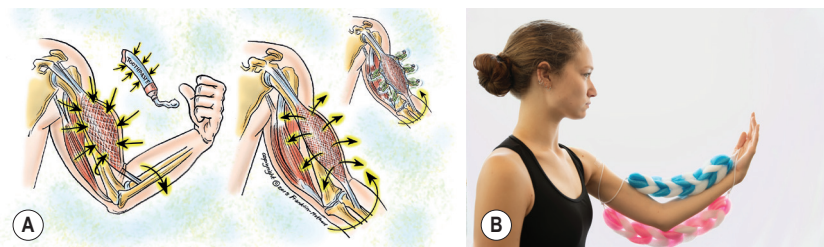


Figure 47.3

Exercise 1: Imaging reciprocal motion of the biceps and triceps fascia of the arm.

(A) DNI anatomical and metaphorical illustration (drawn by Mr. Eric Franklin, all rights reserved), and (B) visual and kinesthetic modeling with sponges (photographer: Karina Glikin-Yaari, all rights reserved).

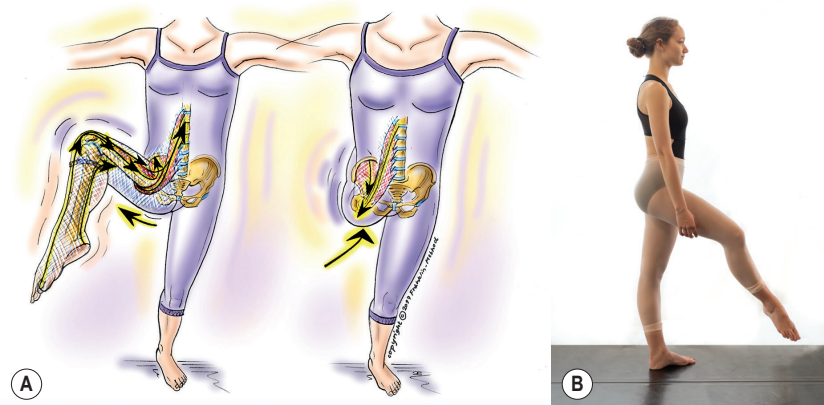


Figure 47.4

Exercise 2: Fascial mobility during a leg motion. (A) DNI illustration of fascial movement during a leg swing (drawn by Mr. Eric Franklin, all rights reserved), and (B) a model demonstrating a leg swinging forward (photographer: Karina Glikin-Yaari, all rights reserved).

differences? Do you feel the right arm more relaxed, elongated and energetic?

11. Repeat the exercise on the left arm.

Exercise 2. Fascial envelope during a leg swing (Figure 47.4)

1. Stand on one leg and swing the other leg backwards and forwards with the thigh lifted and the knee bent.
2. As the leg swings to the front, image the fascia of the posterior aspect of the lower extremity (all the way to the toes) lengthening. Try to image its dynamic connections with the thoraco-lumbar fascia. Image the hip flexors and specifically the iliopsoas muscles shortening and bulging against its fascia (Figure 47.4A).
3. When swinging the leg backward, image the pushing and bulging of the gluteus muscle and erector spinae muscles against their fasciae. Image also the fascia of the anterior aspect of the leg (all the way to the toes) lengthening. Focus specifically on the fascia of the iliopsoas muscle lengthening (Figure 47.4B).
4. You can also image the fascial envelope around the whole body and notice how it changes during the movement. As the leg swings forward, image the posterior fascia with a metaphor of a big net stretching across the wall of your whole back and leg. As the leg swings backward, image the metaphorical net of the anterior fascia lengthening across the front of the body, particularly on the side of the extending leg.

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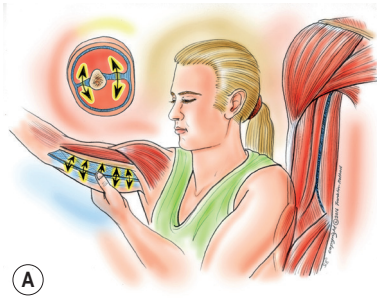


Figure 47.5

Exercise 3: Anatomical and metaphorical imagery of the inter-muscular septa of the arm. **(A)** DNI illustration of palpating the medial inter-muscular septum (drawn by Mr. Eric Franklin, all rights reserved); **(B)** mobilizing the medial and lateral inter-muscular septa (photographer: Karina Glikin-Yaari, all rights reserved).

5. Pause and compare sides. Do you feel any difference? Do you feel one leg is more vivid and elastic?
6. Now perform the same exercise on the other leg.

Exercise 3. Anatomical and metaphorical imagery of the inter-muscular septa of the arm (Figure 47.5)

1. Place your left hand on top of the right biceps muscle with your thumb on the medial intermuscular septum and your fingers on the lateral intermuscular septum, so that your hand is on top of the biceps.
2. Image the space between the biceps and the triceps muscles; this is where the septum is located and the humerus bone can be felt. Now try to palpate it with your fingers, carefully as these areas contain nerves and vessels, and the area may be tender.
3. Rotate your arm medially and laterally, and use your other hand to pull the biceps muscle in the opposite direction. Mobilize both medial and lateral septa, and image them being stretchy like rubber.
4. Now switch to holding the triceps muscle with fingers and thumb on both septa and rotate your arm counter to your gentle pull on the triceps. Image stretchy septa (Figure 47.5A).
5. Move your left fingers down the arm along the septa to the medial and lateral epicondyles of the humerus. As you rub and mobilize the area, image the fascial tissue of the septa below. Also, image the anatomical fascial boundary between the biceps muscle on top of the humerus and the triceps muscle beneath it.
6. Gently rub both septa just proximal to the medial and lateral epicondyles. Then push outward on both epicondyles of humerus. Image the septa lengthening as if you were pulling on a rubber band in a variety of directions (Figure 47.5B).
7. Let go of both hands and notice any differences between the right and left arms. Does the right arm feel more relaxed or energetic? Flex both elbows and notice any differences between them in power and ease of movement.
8. Now switch hands and repeat the process on the other arm. Notice any differences between the two sides.

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Conclusion

MI is a dynamic, complex phenomenon affected by and interacting with various elements, including sensory stimuli from within the body as well as from the environment. The relationship suggested in this chapter between MI and fascia relies on mutual and interactive cognitive and anatomical-physiological mechanisms. As MI of fascial tissue holds potential for enhancing fascial awareness and perception, including it in therapy (e.g., bodywork, physical and occupational therapy, osteopathy, structural integration, etc.), may result in positive outcomes in movement, body schema and pain reduction. Complementary to that, by potentially modulating sensory information from and about the fascia, MI may affect actual

fascia-related sensory and motor experiences. These mechanisms of effect are currently only science-driven hypotheses, but ones which open a window to new research paths. If confirmed, proven connections between MI and fascia could enhance collective knowledge about fascial movement, awareness and perception.

Insofar as MI is a creative process of self-discovery and given the effect of movement on it, the more one moves, the more it is possible for the brain to form and offer mental images and metaphors. Respecting individual differences among movers as well as trial and error is encouraged when it comes to MI. If one image does not work, try a different one. You may be surprised by the results.

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